

channels, and if that playback comprises of more than one digital sound sample, usually there is a software level down mix involved. Sound cards that are bundled on the motherboard mostly work this way.

But some modern day sound cards do provide actual hardware polyphony and are referred to as ‘hardware audio accelerators’. These cards are capable of more than actual voice polyphony and support hardware acceleration of 3D sound, positional audio and real time digital signal processing. They are mainly used for voice mixing and effects processing in hardware, eventually leaving digital filtering and frequency domain conversions for a dedicated DSP. The latest sound cards support upto 8 physical audio channels in the 7.1 speaker setup.

PERSONAL AUDIO

In the vast world that we began to describe in the last section, a good niche to focus on would be the world of personal audio technology. Because it is in this space that we need to make most of our decisions. And at times those decisions are uninformed. But that is going to be a thing of the past soon!

Headphones

You may call it a headphone, ear-phone, or even an ear monitor, but essentially headphones are different from typical speakers only in design. They are designed to be much smaller and private in their functionality, in contrast to loudspeaker that emit the sound into the air. And it is their overall design which you can classify them by. Circumaural or supra-



A circumaural Sennheiser HD555.

aural headphones use a band of plastic or other material to hold the speakers in place by passing over the head. And the other type, known as earbuds or earphones is made of individual units that are plugged into the user's ear canals. Both designs come with wired as well as wireless variants that connect via bluetooth or other wireless technology. The wired variants mainly come with a 3.5mm or a 6.5mm telephone jack. You might wonder why the term ‘telephone’ is used here. The initial headphones, which were the first type of speakers to be used, were mainly used by telephone operators to